

Probability Lecture Notes 3: Distributions and Applications

1. Random Variables

A random variable assigns a numerical value to the outcome of a random experiment.

A discrete random variable takes countable values, such as the number shown on a die.

2. Probability Distribution

A probability distribution shows the probability of each possible value of a random variable.

For a fair six-sided die, each value 1, 2, 3, 4, 5 and 6 has probability $1/6$.

The probabilities in a complete probability distribution must add up to 1.

3. Expected Value

Expected value is the long-run average value expected if an experiment is repeated many times.

$$E(X) = \text{Sum of } [x \times P(x)]$$

For a fair die:

$$E(X) = 1(1/6) + 2(1/6) + 3(1/6) + 4(1/6) + 5(1/6) + 6(1/6) = 3.5$$

The value 3.5 does not mean that 3.5 can be rolled. It is the long-run average.

4. Real-Life Applications

Weather: estimating the chance of rain, storms or other conditions.

Medicine: studying treatment success, risk and test results.

Insurance: estimating the likelihood of claims and accidents.

Finance: analysing uncertainty and investment risk.

Sports: estimating possible match or performance outcomes.

Engineering: estimating reliability and system failure.

Artificial intelligence: making predictions when information is uncertain.

5. Project Example: Coloured Balls

A bag contains 5 red balls, 3 blue balls and 2 green balls. There are 10 balls in total.

$$P(\text{Red}) = 5/10 = 1/2$$

$$P(\text{Blue}) = 3/10$$

$$P(\text{Green}) = 2/10 = 1/5$$

Check: $5/10 + 3/10 + 2/10 = 10/10 = 1$. The probabilities of all possible colours add up to 1.

6. Key Rules to Remember

$$0 \leq P(A) \leq 1$$

$$P(S) = 1$$

$$P(A') = 1 - P(A)$$

$$P(A \text{ or } B) = P(A) + P(B) \text{ (mutually exclusive events)}$$

$$P(A \text{ and } B) = P(A) \times P(B) \text{ (independent events)}$$

$$P(A | B) = P(A \text{ and } B) / P(B)$$

7. Conclusion

Probability is the mathematical study of chance and uncertainty. It helps us measure how likely events are and supports better decisions in situations where outcomes are not certain.

The most important ideas are sample spaces, events, basic probability, complements, addition and multiplication rules, independent and dependent events, conditional probability, probability distributions and expected value.

8. Quick Review

1. Why must the probabilities in a complete distribution add up to 1?
2. What does expected value represent?
3. Give two real-life uses of probability.